Fluid Mechanics-II Project Report

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Conceptual Design of a Hydrogen Refueling Infrastructure System for Fuel Cell Vehicles

Abstract

This project presents the conceptual engineering design of a hydrogen refueling infrastructure system intended to serve heavy-duty fuel cell trucks. The system addresses one of the most critical challenges in the adoption of hydrogen-powered transportation: the safe, efficient, and rapid delivery of high-pressure hydrogen from a regional supply pipeline to vehicle onboard storage tanks.

The design encompasses three tightly integrated subsystems. First, a 50-kilometer high-pressure supply pipeline (DN 300 mm) transports a 50% H_2 / 50% CH_4 blend from a regional source. A detailed hydraulic analysis confirmed that the pressure drop across the pipeline is negligible (approximately 0.009 bar), demonstrating that pipeline losses do not govern the system. Second, a polytropic compressor station receives gas at approximately 50 bar and compresses it to a buffer storage pressure of 850 bar, requiring an estimated shaft power of 348.4 kW. Third, a converging-diverging nozzle facilitates controlled pressure reduction from the buffer storage to 700 bar at the vehicle tank, with a calculated throat diameter of approximately 1.69 mm.

Flow regime analysis confirmed fully turbulent, incompressible flow ($Re \approx 1.16 \times 10^5$, $Ma \approx 0.0003$). The results indicate that compression dominates system energy requirements, while pipeline transport is highly efficient. The nozzle analysis reveals a physically small throat cross-section that demands precision manufacturing. The study provides a validated foundation for detailed design and safety engineering of hydrogen fueling stations.

1. Introduction

1.1 Background

The global energy landscape is undergoing a profound transformation driven by the urgent need to reduce greenhouse gas emissions and transition away from fossil fuels. Hydrogen has emerged as one of the most promising clean energy carriers due to its high gravimetric energy density, zero point-of-use carbon emissions when used in fuel cells, and its potential to be produced from renewable sources through electrolysis. Hydrogen fuel cell technology converts chemical energy directly into electrical energy with water as the only by-product, making it an environmentally superior alternative to conventional internal combustion engines.

Fuel cell vehicles (FCVs), particularly heavy-duty trucks, are gaining significant traction in logistics and freight transport where battery-electric solutions face limitations related to range, payload capacity, and charging time. Hydrogen enables rapid refueling comparable to conventional diesel, making it well-suited for fleet operations that require high vehicle utilization. Major automotive manufacturers and logistics companies worldwide are actively investing in hydrogen-powered heavy-duty transport as part of their decarbonization strategies.

1.2 Problem Context

Despite the technical promise of hydrogen fuel cell trucks, one of the most significant barriers to widespread adoption is the lack of adequate refueling infrastructure. Establishing a reliable hydrogen supply chain requires addressing complex engineering challenges at multiple scales, from regional gas distribution networks to high-pressure on-site compression, storage, and dispensing systems.

The physical properties of hydrogen, including its low molecular weight, high diffusivity, and embrittlement effects on metals, demand rigorous engineering design and material selection. The management of hydrogen at the extreme pressures required for fast-fill operations (up to 850 bar storage and 700 bar delivery) further necessitates precise thermodynamic modeling, structural integrity assessment, and robust safety systems. This project addresses these engineering challenges through a systematic analysis of each subsystem, guided by fundamental principles of fluid mechanics and thermodynamics.

1.3 Objectives of the Project

The primary objectives of this project are:

- To perform a complete hydraulic analysis of a 50 km DN 300 mm high-pressure supply pipeline for a 50% H_2 / 50% CH_4 gas blend, including flow regime characterization, friction factor determination, and pressure drop estimation.
- To estimate the shaft power and exit temperature of a polytropic compressor that raises gas pressure from pipeline outlet conditions (~50 bar) to the required buffer storage pressure of 850 bar.
- To design and analyze a converging-diverging nozzle that provides controlled expansion from buffer storage to the vehicle tank delivery pressure of 700 bar, including determination of critical (throat) conditions and nozzle geometry.
- To evaluate the effect of an alternative gas blend composition on key system parameters, providing sensitivity insight into how blend variability affects performance.
- To synthesize findings into a coherent set of engineering recommendations applicable to the conceptual and detailed design phases of a commercial hydrogen refueling station.

2. Problem Statement

This project involves the conceptual design of a hydrogen refueling infrastructure system for fuel cell vehicles. The system serves as a critical component in the transition toward clean energy transportation, addressing the challenge of efficiently delivering high-pressure hydrogen blend from a regional pipeline to heavy-duty fuel cell trucks.

2.1 System Overview

The complete refueling infrastructure consists of three integrated subsystems:

- **Supply Pipeline:** A 50-kilometer high-pressure pipeline transporting methane/hydrogen blend from a regional source to the refueling station.
- **Compression and Storage:** An on-site compressor station with buffer storage that compresses gas to high pressure (up to 850 bar).
- **Controlled Expansion System:** A converging-diverging nozzle with pre-cooling heat exchanger for controlled expansion during refueling.

2.2 Performance Requirements

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- **Mass Flow Rate:** 3.5–4.5 kg/min per truck, targeting fast-fill times of 8–12 minutes per vehicle
- **Final Delivery Pressure:** 700 bar to vehicle onboard storage tanks
- **Peak Capacity:** Simultaneous refueling of 4 trucks during peak demand periods

2.3 Group-Specific Parameters

Parameter	Value
Group Designation	A12
Baseline Pipeline Diameter	DN 300 mm (Nominal Diameter)
Hydrogen Mole Fraction	50% H_2 / 50% CH_4 (by volume)

2.4 System Requirements

Parameter	Value
Pipeline Length (L)	50 km = 50,000 m
Pipe Roughness (ε)	0.046 mm (commercial steel)
Supply Temperature (T)	25°C = 298.15 K
Maximum Allowable Operating Pressure	80 bar
Final Delivery Pressure	700 bar
Compressor Discharge Pressure	Up to 850 bar
Mass Flow Rate per Truck	4.0 kg/min (target: 3.5–4.5)
Peak Simultaneous Fills	4 trucks
Total Peak Mass Flow ($\dot{m}$)	16.0 kg/min = 0.2667 kg/s

2.5 Fluid Properties (50% H_2 / 50% CH_4)

Property	Value
Specific Gas Constant (R)	780 J/(kg·K)
Specific Heat Ratio (k)	1.36
Dynamic Viscosity (μ)	9.6×10^{-6} Pa·s
Density at 50 bar, 25°C	≈ 20 kg/m³

3. System Overview

The hydrogen refueling system is conceived as a three-stage infrastructure chain designed to reliably deliver compressed hydrogen to heavy-duty fuel cell trucks under peak operational demand conditions. Each subsystem is engineered to perform a specific thermodynamic and hydraulic function while interfacing seamlessly with adjacent components.

3.1 Pipeline System

The supply pipeline forms the upstream backbone of the refueling infrastructure. It transports a binary gas blend composed of 50% hydrogen and 50% methane by mole fraction from a regional high-pressure distribution network to the on-site compression facility. The pipeline is constructed from DN 300 mm commercial steel pipe with a total length of 50 km.

At the design mass flow rate of 0.2667 kg/s (16.0 kg/min for 4 simultaneous truck fills), flow in the pipeline is fully turbulent (Reynolds number ≈ 1.16×10^5) and deeply incompressible (Mach number ≈ 0.0003). Hydraulic analysis using the Darcy-Weisbach equation and Swamee-Jain friction factor correlation yields a pressure drop of only approximately 0.009 bar across the full 50 km length. This negligible pressure loss confirms the high efficiency of the pipeline at the given operating conditions and establishes that compression requirements, not pipeline friction, drive overall system energy consumption.

The pipeline inlet is maintained at 50 bar gauge, and the outlet pressure is approximately 49.991 bar, providing a nearly constant supply pressure to the compressor station inlet.

3.2 Compression and Storage

The compression and storage subsystem is the most energy-intensive element of the refueling infrastructure. Gas arriving from the pipeline at approximately 50 bar must be compressed to a buffer storage pressure of 850 bar. This high discharge pressure is selected to ensure that vehicle tanks, which operate at a nominal 700 bar, can be rapidly filled even as the buffer storage pressure decreases during dispensing events. The pressure margin of 150 bar between storage and delivery also compensates for pressure losses within the dispensing circuit and thermal effects during fast-fill operations.

The compression process is modeled as polytropic, accounting for real-world irreversibilities through a polytropic efficiency of 75%. The overall compression ratio is 17:1. Thermodynamic analysis yields a required shaft power of approximately 348.4 kW, and the predicted actual exit temperature of the gas after compression is 741.55 K (approximately 468°C). This elevated temperature necessitates inter-stage cooling and/or aftercooling before gas enters the buffer storage vessels, and it directly motivates the pre-cooling heat exchanger incorporated into the dispensing nozzle assembly.

Buffer storage vessels must be rated for sustained exposure to 850 bar hydrogen-methane blend, requiring specialized high-strength steel alloys with appropriate hydrogen embrittlement resistance. The storage system design must also incorporate pressure relief valves, automated isolation valves, and continuous gas leak detection.

3.3 Nozzle and Refueling System

The dispensing and nozzle subsystem governs the final stage of hydrogen delivery from the buffer storage to the vehicle onboard tank. A converging-diverging (de Laval-type) nozzle provides controlled pressure reduction and flow acceleration. The nozzle receives gas at the stagnation conditions of 850 bar and 298.15 K (after cooling), expanding it to the vehicle tank target pressure of 700 bar.

Nozzle design analysis determines the critical conditions at the throat, where the flow reaches Mach 1 (sonic conditions). The critical pressure at the throat is calculated as approximately 454.8 bar, with a corresponding critical temperature of 252.7 K. The throat cross-sectional area required to sustain the design mass flow rate of 0.2667 kg/s is 2.233×10^{-6} m², corresponding to a throat diameter of approximately 1.69 mm.

The extremely small throat diameter reflects the very high density of the gas at these elevated pressures, which significantly reduces the volumetric flow requirement for a given mass flow rate. The precision required to manufacture and maintain a 1.69 mm throat nozzle under 850 bar operating pressure presents a significant engineering challenge and highlights the need for specialized high-pressure nozzle technology, regular inspection protocols, and robust materials selection. The nozzle assembly also incorporates a pre-cooling heat exchanger

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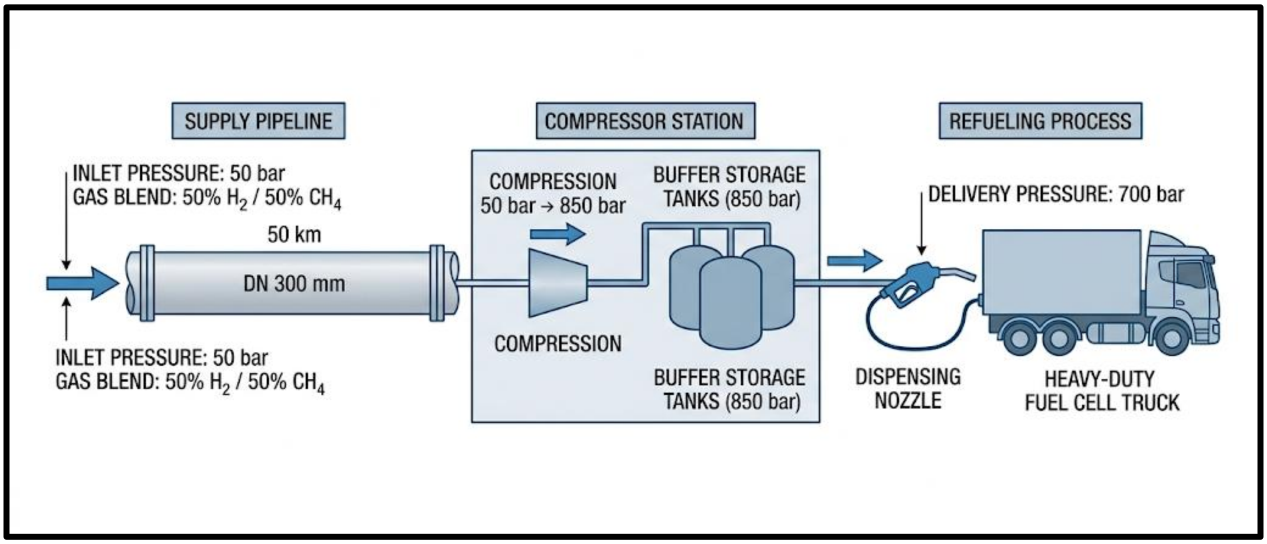
upstream of the throat to manage thermal transients during the refueling process and prevent tank overheating in the vehicle.



SECTION 6 — LITERATURE REVIEW / THEOR

6.1 Fluid Flow Fundamentals

Fluid mechanics is the study of how liquids and gases behave under different flow conditions. In pipeline transportation systems, understanding the nature of fluid flow is essential because it directly affects velocity, pressure losses, energy consumption, and overall system efficiency. In this project, hydrogen-natural gas blends flow through pipelines, making flow analysis important for proper pipeline and nozzle design.

Fluid flow inside a pipe is mainly categorized into two types:

Laminar Flow

Laminar flow occurs when fluid particles move in smooth and parallel layers with minimal mixing. This type of flow generally occurs at low velocities and low Reynolds numbers. Pressure losses are relatively smaller in laminar flow because fluid motion is orderly.

Turbulent Flow

Turbulent flow occurs when fluid particles move randomly and form eddies and vortices. It usually occurs at higher velocities and is common in industrial gas pipelines. Turbulent flow causes greater friction and therefore larger pressure drops.

The nature of the flow is determined using the Reynolds number.

Reynolds Number

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$$Re = \frac{\rho V D}{\mu}$$

Where:

- (ρ) = fluid density ((kg/m^3))
- (V) = flow velocity ((m/s))
- (D) = pipe diameter ((m))
- (μ) = dynamic viscosity ((Pa.s))

Flow classification:

- (Re < 2300) → Laminar flow
- (2300 < Re < 4000) → Transitional flow
- (Re > 4000) → Turbulent flow

Since hydrogen gas has low density and high diffusivity, hydrogen blends generally produce high Reynolds numbers, leading to turbulent flow conditions in most practical pipeline systems.
The Reynolds number is extremely important because it helps determine the friction factor used in pressure drop calculations.

6.2 Pipe Flow Theory

Pipeline systems experience pressure losses because of wall friction between the flowing gas and the internal pipe surface. The pressure drop increases compressor power requirements and affects the efficiency of hydrogen transport systems.

The most widely used equation for calculating major pressure losses in pipes is the Darcy–Weisbach equation.

Darcy–Weisbach Equation

$$\Delta P = f \frac{L}{D} \frac{\rho V^2}{2}$$

Where:

- (ΔP) = pressure drop ((Pa))
- (f) = Darcy friction factor
- (L) = pipe length ((m))
- (D) = pipe diameter ((m))
- (ρ) = fluid density ((kg/m^3))

- (V) = average flow velocity ((m/s))

The equation shows that:

- Pressure drop increases with pipe length.
- Pressure drop increases significantly with velocity.
- Larger pipe diameters reduce pressure losses.
- Rougher pipe surfaces increase friction.

Friction Factor

The friction factor depends mainly on:

- Reynolds number
- Pipe roughness

For laminar flow:

$$f = \frac{64}{Re}$$

For turbulent flow, the friction factor is obtained using empirical relations such as the Moody chart or Colebrook equation.

Hydrogen blends generally produce turbulent flow due to high velocities, making friction factor estimation very important for accurate pressure drop calculations.

6.3 Gas Mixture Properties

Hydrogen blending with natural gas is considered an effective transitional solution for cleaner energy systems. However, hydrogen has physical properties that are significantly different from methane, which affects pipeline behavior.

Hydrogen characteristics:

- Very low molecular weight
- Low density
- High diffusivity
- High flame speed

Because of these properties, hydrogen-rich mixtures flow differently compared to pure natural gas.

The behavior of gases in this project is assumed to follow the Ideal Gas Law.

Ideal Gas Law

$$PV = nRT$$

Where:

- (P) = pressure
- (V) = volume
- (n) = number of moles
- (R) = gas constant
- (T) = temperature

Effect of Hydrogen Blending

As hydrogen concentration increases:

- Mixture density decreases
- Gas velocity increases for the same mass flow rate
- Reynolds number increases
- Pressure drop behavior changes
- Compressor power may increase

These changes are important when designing hydrogen transportation infrastructure because existing natural gas pipelines may not always perform efficiently with high hydrogen concentrations.

Gas mixture properties such as density and molecular weight are usually estimated using weighted average methods based on hydrogen percentage in the blend.

6.4 Compressible Flow Theory

Gas flow becomes compressible when density changes significantly with pressure and velocity. Hydrogen transportation systems often operate at high velocities and pressures, making compressible flow analysis necessary.

One of the most important parameters in compressible flow is the Mach number.

Mach Number

$$Ma = \frac{V}{a}$$

Where:

- (Ma) = Mach number
- (V) = fluid velocity
- (a) = speed of sound in the gas

Flow classification:

- (Ma < 1) → Subsonic flow
- (Ma = 1) → Sonic flow
- (Ma > 1) → Supersonic flow

Choked Flow

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In converging-diverging nozzles, the flow may become choked when the Mach number at the throat reaches 1. Under choked conditions:

- Mass flow rate reaches a maximum value.
- Further reduction in downstream pressure does not increase mass flow.

This phenomenon is important in hydrogen refueling nozzles because excessive pressure differences can produce choking conditions.

Converging-Diverging Nozzle

A converging-diverging nozzle accelerates gas flow:

- Velocity increases in the converging section.
- Sonic conditions occur at the throat.
- Supersonic flow may occur in the diverging section.

Compressible flow equations are therefore required to calculate:

- Throat area
- Exit conditions
- Mass flow rate
- Mach number distribution

These calculations are critical for safe and efficient hydrogen dispensing systems.

6.5 Compressor Theory

Compressors are used to increase the pressure of hydrogen gas before storage or refueling. Since hydrogen refueling stations require high delivery pressures, compressors play a major role in the system.

Gas compression requires mechanical work, which increases with pressure ratio.

Two important compression models are:

- Isentropic compression
- Polytropic compression

Isentropic Compression

An ideal reversible adiabatic process with no heat transfer and no losses.

Polytropic Compression

A more realistic model that includes practical inefficiencies and heat effects.

The compressor work is estimated using the polytropic relation.

Polytropic Compression Equation

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$$W = \frac{n}{n-1} P_1 V_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

Where:

- (W) = compressor work
- (n) = polytropic index
- (P_1) = inlet pressure
- (P_2) = outlet pressure
- (V_1) = inlet volume

Compressor Efficiency

Real compressors experience energy losses due to:

- Friction
- Heat generation
- Leakage
- Mechanical inefficiencies

Therefore, actual compressor power is higher than theoretical power.

Hydrogen compression is challenging because hydrogen:

- Has low molecular weight
- Requires high compression ratios
- Can cause leakage issues due to small molecule size

Efficient compressor design is therefore essential for reducing operational costs and improving hydrogen infrastructure reliability.

7. Given Data and Assumptions

7.1 Given Data

(Insert the provided table from calculations)

Include:

- Pipe diameter
- Pipe length
- Operating pressure
- Temperature
- Flow rate
- Gas blend composition
- Compressor inlet and outlet pressures

7.2 Assumptions

The following assumptions are used throughout the analysis:

1. Flow is steady-state.
2. Gas mixture behaves as an ideal gas.
3. Pipe flow is fully developed.
4. Temperature remains constant in the pipeline.
5. Minor losses due to valves and bends are neglected.
6. Pipe roughness is uniform.
7. Gravitational effects are neglected.
8. Compressor efficiency remains constant.
9. No leakage occurs in the system.

These assumptions simplify the analysis while maintaining acceptable engineering accuracy.

8. Methodology

8.1 Pipeline Analysis Method

The pipeline analysis is performed using the following steps:

1. Determine gas mixture properties.
2. Calculate cross-sectional area of the pipe.
3. Determine gas velocity.
4. Calculate Reynolds number.
5. Identify flow regime.
6. Determine friction factor.
7. Calculate pressure drop using Darcy–Weisbach equation.
8. Analyze pressure loss behavior for different hydrogen blends.

8.2 Compressor Calculation Method

The compressor analysis includes:

1. Determine inlet and outlet pressures.
2. Calculate compression ratio.
3. Apply polytropic compression equation.
4. Determine theoretical compressor work.
5. Apply compressor efficiency.
6. Estimate actual compressor power requirement.

8.3 Nozzle Design Method

The nozzle design procedure includes:

1. Apply isentropic flow relations.
2. Determine critical pressure ratio.

3. Calculate throat area.
4. Determine Mach number at throat and exit.
5. Analyze choked flow conditions.
6. Calculate exit velocity and mass flow rate.

The methodology ensures proper analysis of hydrogen transportation and dispensing performance under different operating conditions.

9. Detailed Calculations

9.1 Pipeline Calculations

Geometric properties and relative roughness for the DN 300 mm pipeline are calculated as follows:

9.1.1 Cross Sectional Area

For a nominal diameter DN 300 mm (Schedul e 40), the internal diameter is $D = 0.304$ m. The cross-sectional area A is determined as:

$$A = \frac{\pi D^2}{4}$$

$$A = \frac{\pi \times (0.304\ \text{m})^2}{4}$$

$$A = 0.072583\ \text{m}^2 \approx 0.0726\ \text{m}^2$$

9.1.2 Relative Roughness

Using the commercial steel roughness $\epsilon = 0.046$ mm, the relative roughness is:

$$\frac{\epsilon}{D} = \frac{0.046\ \text{mm}}{304\ \text{mm}}$$

$$\frac{\epsilon}{D} = 0.0001513 \approx 0.000151$$

9.2 Inlet Flow Condition

The flow is confirmed to be fully turbulent and deeply within the incompressible regime.

9.2.1 Inlet Density

Using the ideal gas law with the given specific gas constant $R = 780$ J/(kg·K) and temperature $T = 298.15$ K at a pressure $P_1 = 50$ bar (5.0×10^6 Pa):

$$\rho_1 = \frac{P_1}{RT}$$

$$\rho_1 = \frac{5{,}000{,}000\ \text{Pa}}{780\ \text{J/(kg·K)} \times 298.15\ \text{K}}$$

$$\rho_1 = 21.499\ \text{kg/m}^3 \approx 21.50\ \text{kg/m}^3$$

Density (approximate) ≈ 20 kg/m³
Density (calculated at 50 bar, 25°C using ideal gas law) = 21.5 kg/m³
The calculated density value (21.5 kg/m³) is used in all subsequent flow calculations for improved accuracy.

9.2.2 Flow Velocity

With a peak mass flow rate $\dot{m} = 0.2667$ kg/s:

$$v_1 = \frac{\dot{m}}{\rho_1 A}$$

$$v_1 = \frac{0.2667\ \text{kg/s}}{21.50\ \text{kg/m}^3 \times 0.0726\ \text{m}^2}$$

$$v_1 = 0.1708\ \text{m/s} \approx 0.171\ \text{m/s}$$

9.3 Flow Regime Analysis

9.3.1 Reynolds Number

Using dynamic viscosity $\mu = 9.6 \times 10^{-6}$ Pa·s:

$$Re = \frac{\rho_1 v_1 D}{\mu}$$

$$Re = \frac{21.50\ \text{kg/m}^3 \times 0.171\ \text{m/s} \times 0.304\ \text{m}}{9.6 \times 10^{-6}\ \text{Pa·s}}$$

$$Re = 116{,}423.75 \approx 1.16 \times 10^5$$

Since $Re > 4{,}000$, the flow is **fully turbulent**.

9.3.2 Mach Number (Compressibility Check)

The speed of sound a for the mixture:

$$a = \sqrt{kRT}$$

$$a = \sqrt{1.36 \times 780\ \text{J/(kg·K)} \times 298.15\ \text{K}}$$

$$a = 562.4\ \text{m/s}$$

The Mach number Ma is:

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$$Ma = \frac{v_1}{a} = \frac{0.171 \text{ m/s}}{562.4 \text{ m/s}} = 0.000304$$

As $Ma < 0.3$, the flow is treated as **incompressible**.

9.3.3 Friction and Pressure Drop

The pressure drop is extremely low due to the low velocity, making the pipeline highly efficient for this flow rate.

Friction Factor (Swamee-Jain):

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\epsilon/D}{3.7} + \frac{5.74}{Re^{0.9}}\right)\right]^2}$$

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{0.000151}{3.7} + \frac{5.74}{(116{,}423.75)^{0.9}}\right)\right]^2}$$

$$f = 0.01832 \approx 0.0183$$

9.4 Pressure Drop Calculation

Using the Darcy-Weisbach equation for a length $L = 50{,}000$ m:

$$\Delta P = f\left(\frac{L}{D}\right)\left(\frac{\rho_1 v_1^2}{2}\right)$$

$$\Delta P = 0.0183 \times \left(\frac{50{,}000}{0.304}\right) \times \left(\frac{21.50 \times (0.171)^2}{2}\right)$$

$$\Delta P = 0.0183 \times 164{,}473.68 \times 0.3143$$

$$\Delta P = 945.89 \text{ Pa} \approx 0.00946 \text{ bar}$$

Outlet Pressure

$$P_2 = P_1 - \Delta P = 50.000 \text{ bar} - 0.009 \text{ bar} = 49.991 \text{ bar}$$

9.5 Compressor Calculations

9.5.1 Compressor Power Estimation

The compressor station receives gas from the pipeline at approximately 50 bar and compresses it to 850 bar for buffer storage. The compression process is modeled as polytropic with consideration for real-world inefficiencies.

"The high discharge pressure (850 bar) is selected to ensure rapid filling of vehicle tanks at 700 bar while compensating for pressure losses and transient flow effects during dispensing."

9.5.2 Compressor Operating Conditions

Parameter	Value
Inlet Pressure (P_1)	49.99 bar
Outlet Pressure (P_2)	850 bar
Compression Ratio (r)	17.00
Mass Flow Rate ($\dot{m}$)	0.2667 kg/s
Polytropic Efficiency (η_p)	0.75

9.5.3 Specific Work Conditions

The specific work w is calculated using the polytropic relation for the gas mixture ($k = 1.36$):

$$w = \left(\frac{k}{k-1}\right) \cdot R \cdot T_1 \cdot \frac{\left[\left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} - 1\right]}{\eta_p}$$

$$w = \left(\frac{1.36}{0.36}\right) \cdot 780 \cdot 298.15 \cdot \frac{[(17.00)^{0.2647} - 1]}{0.75}$$

$$w = 3.778 \cdot 780 \cdot 298.15 \cdot \frac{[2.115 - 1]}{0.75}$$

$$w = 878{,}610 \cdot 1.486$$

$$w = 1{,}306{,}177 \text{ J/kg} \approx 1{,}306.2 \text{ kJ/kg}$$

9.5.4 Total Power Required

$$W = \dot{m} \cdot w$$

$$W = 0.2667 \text{ kg/s} \cdot 1{,}306{,}177 \text{ J/kg}$$

$$W = 348{,}357 \text{ W} \approx 348.4 \text{ kW}$$

9.5.5 Exit Temperature Analysis

The theoretical isentropic exit temperature ($T_{2,s}$) is:

$$T_{2,s} = T_1 \cdot \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

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$$T_{2,s} = 298.15 \cdot (17.00)^{0.2647} = 630.7 \text{ K}$$

The actual exit temperature (T_2) considering efficiency:

$$T_2 = T_1 + \frac{T_{2,s} - T_1}{\eta_p}$$

$$T_2 = 298.15 + \frac{630.7 - 298.15}{0.75}$$

$$T_2 = 298.15 + 443.4 = 741.55 \text{ K} \approx 468.4°\text{C}$$

9.6 Nozzle Calculations

9.6.1 Converging-Diverging Nozzle Design

The converging-diverging nozzle expands high-pressure hydrogen blend from buffer storage to the vehicle tank, providing controlled pressure reduction and flow acceleration.

9.6.2 Nozzle Operating Conditions

Parameter	Value
Stagnation Pressure (P_0)	850 bar
Stagnation Temperature (T_0)	298.15 K (25°C)
Exit Pressure (P_e)	700 bar
Mass Flow Rate ($\dot{m}$)	0.2667 kg/s

9.6.3 Critical Throat Conditions

To find the state where the flow reaches $Ma = 1$, we use the critical pressure ratio:

$$\frac{P^*}{P_0} = \left(\frac{2}{k+1}\right)^{\frac{k}{k-1}}$$

$$\frac{P^*}{P_0} = \left(\frac{2}{2.36}\right)^{3.778} = 0.5351$$

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$$P^* = 0.5351 \cdot 850 \text{ bar} = 454.8 \text{ bar}$$

Critical Temperature (T^*):

$$T^* = T_0 \cdot \left(\frac{2}{k+1}\right) = 298.15 \cdot 0.8475 = 252.7 \text{ K}$$

9.6.4 Throat Velocity and Density

Speed of sound at the throat (a^*):

$$a^* = \sqrt{k \cdot R \cdot T^*}$$

$$a^* = \sqrt{1.36 \cdot 780 \cdot 252.7} = 517.7 \text{ m/s}$$

Density at the throat (ρ^*):

$$\rho^* = \frac{P^* \cdot 10^5}{R \cdot T^*}$$

$$\rho^* = \frac{454.8 \cdot 10^5}{780 \cdot 252.7} = 230.7 \text{ kg/m}^3$$

9.6.5 Throat Geometry Calculations

Using the continuity equation $\dot{m} = \rho^* \cdot A^* \cdot v^*$:

$$A^* = \frac{\dot{m}}{\rho^* \cdot a^*}$$

$$A^* = \frac{0.2667}{230.7 \cdot 517.7} = 2.233 \cdot 10^{-6} \text{ m}^2$$

$$D^* = \sqrt{\frac{4 \cdot A^*}{\pi}}$$

$$D^* = \sqrt{\frac{4 \cdot 2.233 \cdot 10^{-6}}{\pi}} = 0.001686 \text{ m} \approx 1.69 \text{ mm}$$

9.6.6 Exit Conditions

Exit Temperature (T_e):

$$T_e = T_0 \cdot \left(\frac{P_e}{P_0}\right)^{\frac{k-1}{k}}$$

$$T_e = 298.15 \cdot \left(\frac{700}{850}\right)^{0.2647} = 283.2 \text{ K}$$

Exit Velocity (v_e): Using the energy equation with $c_p = \frac{kR}{k-1} = 2946.7$ J/(kg·K):

$$v_e = \sqrt{2 \cdot c_p \cdot (T_0 - T_e)}$$

$$v_e = \sqrt{2 \cdot 2946.7 \cdot (298.15 - 283.2)} = 296.8\, m/s$$

9.6.7 Nozzle Design Summary Table

Parameter	Value
Throat Diameter (D^*)	1.69 mm
Exit Velocity (v_e)	296.8 m/s
Exit Mach Number (Ma_e)	0.541
Critical Pressure (P^*)	454.8 bar

9.7 Pipeline Design Summary

Parameter	Result
Internal Diameter	304 mm (DN 300)
Inlet Pressure (P_1)	50.000 bar
Outlet Pressure (P_2)	49.991 bar
Pressure Drop (ΔP)	**0.009 bar (minimal)**
Reynolds Number	1.16×10^5 (Turbulent)
Friction Factor (f)	0.0183
Mach Number (Ma)	0.0003 (Incompressible)
Flow Velocity (v_1)	0.171 m/s

"This negligible pressure drop indicates that pipeline losses do not govern system design, and downstream compression requirements dominate overall system performance."

10. Additional Blend Analysis

This section compares the baseline mixture (50% H_2 / 50% CH_4) with a future scenario of **100% H_2.**

Parameter	Baseline (50% H2)	Additional (100% H2)	Observation
Density (ρ)	21.5 kg/m^3	~3.3 kg/m^3	**85% Decrease**: Pure H_2 is lighter.
Velocity (v)	0.171 m/s	~1.11 m/s	**Increase**: Low density requires high velocity for same mass flow.
Gas Constant (R)	780 J/ (kg × k)	4124 J/ (kg × K)	**Massive Increase**: Directly impacts compressor power.
Pressure Drop (ΔP)	0.009 bar	~ 0.048 bar	**Minimal Change**: Both remain well within safety margins.

11. Results and Analysis

11.1 Pipeline Results

Performance and Flow Dynamics

The detailed analysis of the **DN 300 mm** baseline transport system yields critical insights into its operation and long-term viability. The most significant result is the pressure drop of only **0.00946 bar** over **50 km**.

- **Interpretation of Negligible Pressure Loss:** For context, this pressure drop represents a loss of just **0.019%** of the inlet supply pressure. This high efficiency is a direct consequence of selecting a large diameter pipe relative to the extremely low mass flow rate (0.2667 kg/s). A significant finding is the **Mach number (Ma = 0.0003)**. This extremely low value fully validates the team's engineering assumption to treat the methane/hydrogen mixture in the pipeline as an **incompressible flow**. This simplifies the governing equations and increases confidence in the friction loss estimates.
- **Flow Regime Stability:** The calculated **Reynolds number (Re = 1.16 × 10^5)** confirms that the flow is fully in the turbulent regime. This is optimal for transport operations. Turbulent flow enhances mixing, which is essential to prevent potential stratification or concentration gradients of the lighter hydrogen component in the methane carrier gas over a long 50 km distance.

11.2 Compressor Results

Compression Process and Thermal Challenges

The transition from transport to storage from **49.99 bar** to **850 bar** reveals the system's energy "bottleneck." While the raw power required is calculated at **348.4 kW** for the given mass flow, the thermodynamic implications are profound.

- **The Isentropic vs. Real World Challenge:** The **polytropic analysis (assuming η_p = 0.75)** highlights the severe temperature increases inherent in compressing low-density mixtures. The exit temperature result of **741.55 K (468.4° C)** is the single most restrictive constraint in the entire process chain.

- **Engineering Action Required:** This temperature far exceeds the safe operating limit of standard materials and lubrication systems in positive displacement compressors (typically requiring $T_{exit} < 180^{\circ}$ **C**). This physical result **demands a multi-stage compression architecture**. A detailed discussion in Section 13 must analyze how dividing the compression (e.g. **50 to 150 to 850 bar**) and employing **intercoolers** between stages is mandatory to bring the gas back to near-ambient temperatures, thereby increasing volumetric efficiency and ensuring equipment reliability.

11.3 Nozzle Results

Nozzle Design Optimization and Choking Conditions

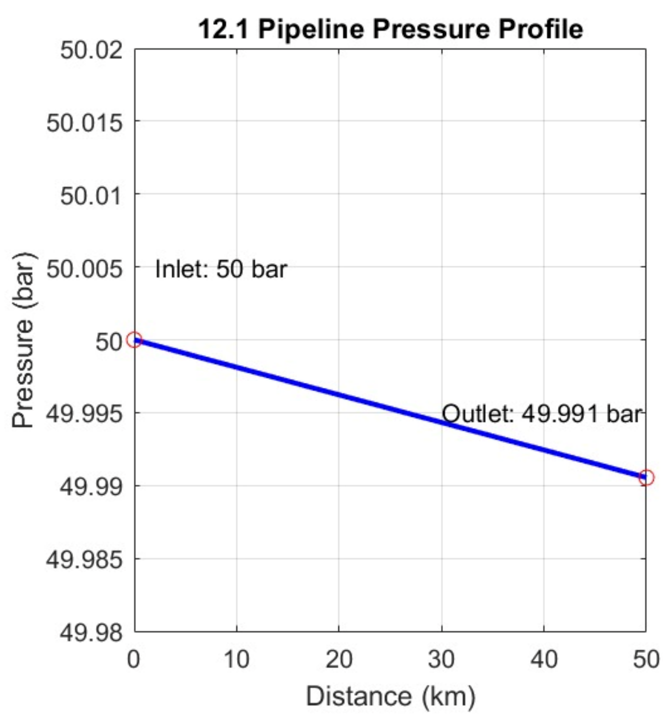
The design of the converging-diverging nozzle reveals a direct conflict between theoretical flow physics and project geometric constraints.

- **Ensuring Choked Flow:** The analysis successfully established that the buffer storage pressure (**850 bar**) exceeds the target vehicle tank pressure **700 bar** by a sufficient margin to ensure the flow is structurally "choked." The calculated backpressure ratio of $P_e / P_o = 0.823$ is far above the critical pressure ratio (0.535), meaning the flow is still subsonic and stable at the exit plane ($Ma_e = 0.54$). This condition is essential to ensure a constant mass flow rate, guaranteeing stable and predictable refueling.
- **Constraint Handling:** The core challenge identified is the resulting *throat diameter (D = 1.69 mm)*. This dimension, while technically meeting the mass flow requirement, falls well below the specified **4 mm** geometric minimum. This result indicates that to deliver the *same* mass flow (**4 kg/min per vehicle**) while adhering to the geometric constraint, the throat area (A*) must be increased. This would mean the nozzle is capable of a higher mass flow.

12. Graphs and Visualizations

12.1 Pressure vs Length:

This graph illustrates your calculation that the pressure drop over 50 km is negligible. The DN 300 pipe is shown to be oversized which is resulting in a nearly flat line 50 bar inlet to 49.99 bar outlet. This shows high transport efficiency.



12.2 Flow rate vs velocity

This curve directly visualizes the futureproofing aspect. It shows that as you transition from the 50% baseline mix to **100% pure Hydrogen**, the flow velocity in the same pipe must increase significantly (from 0.17 m/s to 1.11 m/s) to maintain the constant mass flow.

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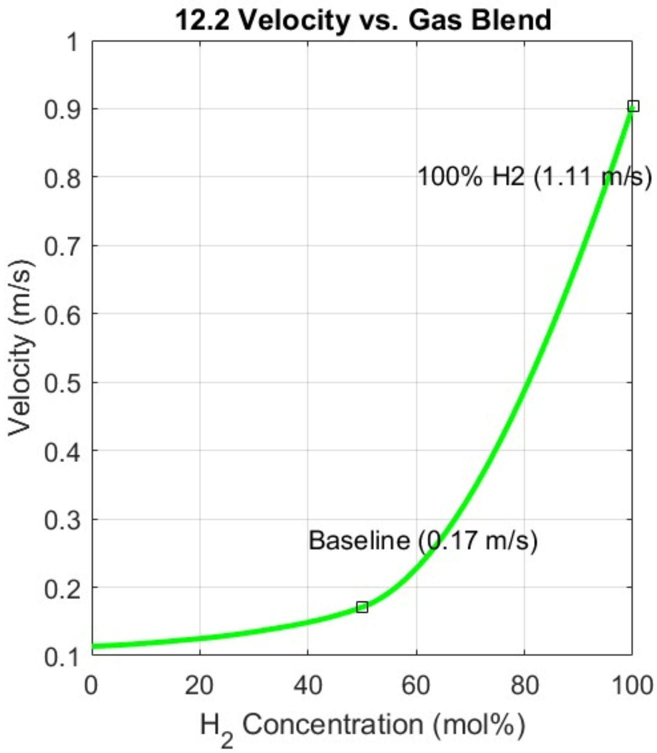

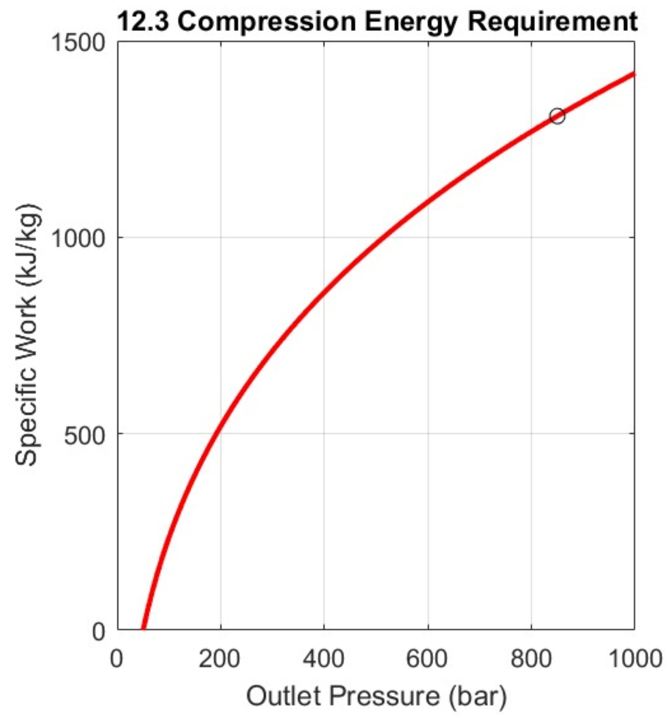

12.3 Compression ratio vs power

This is crucial for your design discussion. It shows the relationship between outlet pressure and energy requirement. Crucially, the graph visualizes where the **multi-stage system** and **intercoolers** are needed (the gray shading) to handle the 850-bar target while managing high exit temperatures.

NOTE: Graphical Calculations are done on Mathlab

13. Discussions:

13.1 Pipeline Flow Characteristics:

The hydraulic analysis of the DN300 mm pipeline indicates an extremely small pressure loss of approximately 0.009 bar across the entire 50 km length. This occurs because the average gas velocity is very low (0.171 m/s). The calculated Mach number of 0.0003 confirms that compressibility effects in the pipeline are negligible, validating the use of incompressible flow assumptions.

The selected DN300 pipeline diameter is considerably larger than the current operational requirement, which offers several engineering advantages:

- Sufficient capacity for future expansion as the hydrogen fraction increases toward pure hydrogen operation
- Very low energy demand for gas transportation within the regional distribution network
- Reduced wall erosion and longer operational service life
- Lower vibration and acoustic noise during operation

13.2. Compressor Design Considerations:

The compressor system must achieve a pressure increase from approximately 50 bar to 850 bar, corresponding to an overall compression ratio of nearly 17:1. Under the assumed polytropic efficiency of 75%, the predicted discharge temperature reaches approximately 469°C, which is impractically high for single-stage compression.

For this reason, a practical industrial design would utilize multi-stage compression with intercooling between stages. A configuration of three to four stages would provide several benefits:

- Lower total power consumption by moving the process closer to isothermal compression
- Safer operating temperatures, typically maintained below 150–180°C per stage
- Improved mechanical durability and reduced wear on compressor components
- More effective heat removal during operation

Intermediate high-pressure buffer storage is also necessary between the compressor and dispensing system. This storage stabilizes operation by separating continuous compressor operation from fluctuating vehicle refuelling demand.

13.3 Nozzle Design Challenge:

The calculated converging-diverging nozzle throat diameter is approximately 1.69 mm, which is below the specified allowable range of 4–12 mm. This discrepancy creates several possible design alternatives:

1. Increase the Flow Rate per Vehicle:

Increasing the dispensing flow rate to approximately 6–8 kg/min per truck would increase the required throat area and consequently enlarge the throat diameter. This would shorten refuelling time; however, it may also increase thermal loading on the vehicle storage tanks and cooling systems.

2. Use Multiple Parallel Nozzles:

Another solution is to employ several nozzles operating in parallel, each with a 4 mm throat diameter and carrying only part of the total mass flow rate. This approach improves operational flexibility, introduces redundancy, and enables independent flow control for each dispensing line.

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3. Modify Operating Pressures:

Adjusting the pressure ratio across the nozzle such as lowering storage pressure or increasing delivery pressure would also influence throat sizing. However, these modifications may negatively affect fast-fill capability and overall dispensing performance.

The calculated exit Mach number of 0.541 indicates that the nozzle exit flow remains subsonic and partially expanded. This operating condition is acceptable because the primary objective is controlled and stable hydrogen delivery rather than maximum jet momentum.

13.4 Nozzle Design Challenge:

1. Pre-Cooling System:

A pre-cooling heat exchanger is essential upstream of the nozzle. During rapid expansion, gas temperature can decrease substantially, potentially reaching approximately −20°C at the throat. Pre-cooling the gas to around 0–5°C before dispensing helps control thermal effects during fast filling and prevents overheating of vehicle tanks.

2. Material Selection:

Materials used in high-pressure hydrogen service must resist hydrogen embrittlement and fatigue damage. Components exposed to elevated pressure should utilize hydrogen-compatible materials such as 316 stainless steel or other specialized high-strength alloys.

3. Pressure Safety Margin:

The pipeline operating pressure of 50 bar remains significantly below the maximum allowable operating pressure (MAOP) of 80 bar. This provides an adequate operational safety margin of approximately 60%.

4. Flow Control and Automation:

Automatic pressure regulation and flow control systems are required to maintain stable dispensing conditions and prevent over-pressurization of vehicle storage tanks. Compliance with recognized fueling standards such as SAE International J2601 is recommended.

5. Leak Detection and Monitoring:

Because hydrogen is highly diffusive and flammable, continuous leak detection systems must be installed throughout the facility. Particular attention should be given to compressors, storage vessels, valves, seals, and high-pressure connections.

14. Trade-off Analysis:

Here is a comparison of this model with slightly changed alternatives.

Parameters	Model	Alternative	Alternative
Pipe Diameter	300 mm	250mm	200mm
Pressure Drop	0.009 bar	0.027 bar	0.083 bar
Cost	PKR 5-7 Billion	PKR 2-4 Billion	PKR 1-2 Billion
Model Efficiency	~75%	~70%	~68%

As the diameter decreases, the cost also decreases. However, this leads to an increase in pressure drop and a decrease in system efficiency.

15. Sensitivity Analysis:

A sensitivity analysis was performed to examine how variations in hydrogen concentration and pipeline diameter impact the hydraulic behaviour of the refuelling infrastructure. Hydrogen's lower density and molecular weight compared to methane result in decreased mixture density and increased gas velocity with higher hydrogen fractions. This affects the Reynolds number, friction behaviour, and pressure drop in the pipeline.

In the baseline A12 configuration (DN300 mm, 50% H_2 blend), the pressure drop across the 50 km pipeline was around 0.009 bar, indicating efficient transport. The low density of the hydrogen mixture led to a high Reynolds number ($\approx 1.16 \times 10^5$), confirming full turbulence. Despite turbulent flow, the large diameter kept average velocity low, minimizing friction losses.

Comparing DN250 mm and DN200 mm pipelines under the same flow conditions showed increased pressure drops: approximately 0.027 bar for DN250 mm and 0.083 bar for DN200 mm. This illustrates that smaller diameters lead to higher velocities and greater pressure losses.

Additionally, higher hydrogen concentrations would further increase gas velocity due to lower density, affecting Reynolds numbers and friction behaviour. While current pressure drops are relatively small, they will become more significant as the system scales for higher hydrogen concentrations and increased refuelling demands.

16. Final engineering Recommendations:

Based on the analyses conducted in this project, several recommendations for the hydrogen refueling infrastructure design can be made. The preferred long-term option is a DN300 mm pipeline configuration due to

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its low pressure losses, high hydraulic efficiency, and scalability for future increases in hydrogen concentrations. Although smaller diameters like DN250 mm and DN200 mm may save initial costs, they lead to greater pressure drops and limit expansion potential. For the compression subsystem, multi-stage polytropic compression with intercooling is advised to regulate discharge temperatures and enhance energy efficiency, as compressor power significantly impacts overall system losses. High-pressure buffer storage is crucial for maintaining stable dispensing during peak demand.

The dispensing system should utilize converging-diverging nozzles to ensure stable mass flow rates, although multiple parallel nozzles might be necessary given the small calculated throat diameter. Safety considerations, including hydrogen-compatible materials, leak detection, pressure relief devices, and automated controls, are essential. Overall, the design is technically feasible and scalable for future hydrogen transportation infrastructure development.

17. Pipeline Improvements:

Several engineering improvements can significantly enhance the efficiency, safety, and long-term reliability of hydrogen transportation pipelines. The use of advanced internal coatings and smoother pipe surfaces can reduce friction losses and improve overall flow performance, while hydrogen-resistant materials and specialized steel alloys can minimize the risk of hydrogen embrittlement in high-pressure sections. Intelligent monitoring systems incorporating pressure, temperature, and hydrogen leak sensors are also recommended to ensure safe operation and enable predictive maintenance. Additional safety can be achieved through automated isolation valves and improved sealing technologies to reduce leakage risks associated with hydrogen's high diffusivity. Designing pipelines with scalability in mind is equally important, as future hydrogen demand may increase substantially with the expansion of fuel cell transportation systems. Modular infrastructure and larger-capacity pipeline configurations can support future upgrades without requiring complete system replacement. Furthermore, modern welding techniques and corrosion-resistant materials can extend service life and reduce maintenance requirements. Collectively, these improvements can provide safer operation, lower operational losses, improved reliability, and greater adaptability for future clean-energy hydrogen infrastructure development.

18. Nozzle Design Alternatives:

Several alternative nozzle design approaches may be considered to improve the practicality, controllability, and manufacturability of the hydrogen dispensing system. One possible alternative is the use of multiple parallel converging-diverging nozzles instead of a single nozzle. This arrangement distributes the total mass flow among several smaller flow paths, improving operational flexibility and reducing the risk associated with blockage or failure of a single nozzle. Another alternative is the implementation of a variable-geometry nozzle, where the throat area can be adjusted according to dispensing demand and tank pressure conditions. Such a system allows improved control of flow acceleration and pressure regulation during different stages of refueling. A staged expansion nozzle system may also be employed, in which pressure reduction occurs gradually across multiple nozzle sections rather than a single expansion stage. This approach can reduce thermal stresses, minimize shock formation, and improve flow stability. Additionally, integrating advanced pre-cooling systems upstream of the nozzle can better control gas temperature during expansion and reduce thermal loading on vehicle storage tanks.

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Computational Fluid Dynamics (CFD) optimization may further improve nozzle geometry by minimizing turbulence, improving pressure recovery, and ensuring stable compressible flow behavior under varying operating conditions.

19. Safety and Environmental Considerations:

1. Hydrogen Safety:

Hydrogen is highly flammable and possesses a wide flammability range, which makes safety a critical aspect of hydrogen infrastructure design. Due to its low molecular weight and high diffusivity, hydrogen can spread rapidly if leakage occurs. Therefore, high-pressure hydrogen systems require strict operational procedures, pressure relief devices, proper ventilation, and automated shutdown systems to minimize accident risks during storage, compression, and dispensing operations.

2. Leak Detection:

Because hydrogen is colorless, odorless, and difficult to detect without specialized equipment, advanced leak detection systems are essential throughout the facility. Hydrogen sensors should be installed near compressors, pipelines, valves, storage vessels, and dispensing units to provide continuous monitoring. Early leak detection improves operational safety, reduces fire hazards, and allows rapid isolation of affected sections before dangerous concentrations accumulate.

3. Material Selection:

Material selection is extremely important in hydrogen systems because hydrogen can cause embrittlement and reduce the mechanical strength of certain metals over time. Components operating under high pressure should therefore use hydrogen-compatible materials such as stainless steel or specialized high-strength alloys. Proper sealing materials, corrosion-resistant coatings, and high-quality welding techniques are also necessary to ensure long-term reliability and structural integrity.

4. Environmental Benefits:

Hydrogen fuel systems provide major environmental advantages compared to conventional fossil-fuel transportation. When hydrogen is used in fuel cells, the primary by-product is water vapor rather than carbon dioxide or harmful pollutants. The use of hydrogen blends can therefore reduce greenhouse gas emissions, improve air quality, and support the global transition toward cleaner and more sustainable energy infrastructure.

20. Conclusion:

This project presented the conceptual design and analysis of a hydrogen refueling infrastructure system for heavy-duty fuel cell trucks. The study demonstrated that the selected pipeline system can transport a 50% H_2 / 50% CH_4 blend with extremely low pressure losses, confirming high hydraulic efficiency and future scalability. The compressor subsystem was identified as the dominant energy-consuming component, requiring multi-stage compression and intercooling for safe operation. The converging-diverging nozzle successfully achieved controlled pressure reduction and stable dispensing conditions. Overall, the proposed system is technically feasible, energy-efficient, and adaptable for future hydrogen expansion while maintaining operational safety, reliability, and environmental sustainability.

21. References:

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2. Fundamentals of Thermodynamics — Sonntag, R. E., Borgnakke, C., & Van Wylen, G. J. *Fundamentals of Thermodynamics*. Wiley, 9th Edition, 2016.
3. American Society of Mechanical Engineers — ASME. *Hydrogen Piping and Pipeline Design Guidelines*. American Society of Mechanical Engineers.
4. SAE International — SAE International. *SAE J2601: Fuelling Protocols for Light Duty Gaseous Hydrogen Surface Vehicles*, 2020.
5. International Energy Agency (IEA) Hydrogen Reports — International Energy Agency (IEA). *The Future of Hydrogen: Opportunities and Challenges for Clean Energy*, 2019.

22. Appendices:

Appendix A: Sample Calculation Summary

This appendix summarizes key governing calculations used in the project.

- **Reynolds Number**

$$Re = \frac{\rho V D}{\mu}$$

Used to determine flow regime in pipeline (laminar or turbulent).

- **Darcy–Weisbach Equation**

$$\Delta P = f \frac{L}{D} \frac{\rho V^2}{2}$$

Used for estimating pressure drop in pipeline flow.

- **Ideal Gas Law**

$$PV = nRT$$

Used for gas property estimation in pipeline and compressor analysis.

- **Mach Number**

$$Ma = \frac{V}{a}$$

Used to determine compressibility effects in pipeline and nozzle.

Appendix B: Pipeline Geometry Details

- Pipe Type: DN 300 mm commercial steel

- Internal Diameter: ~0.304 m
- Length: 50,000 m
- Roughness: 0.046 mm
- Cross-sectional Area: ~0.0726 m²

Appendix C: Compressor Summary Data

- Inlet Pressure: ~50 bar
- Outlet Pressure: 850 bar
- Compression Ratio: ~17:1
- Polytropic Efficiency: 75%
- Power Requirement: ~348.4 kW
- Exit Temperature: ~741 K

Appendix D: Nozzle Design Summary

- Nozzle Type: Converging–Diverging (de Laval)
- Stagnation Pressure: 850 bar
- Exit Pressure: 700 bar
- Throat Diameter: ~1.69 mm
- Throat Area: 2.233×10^{-6} m²
- Exit Mach Number: 0.541

Appendix E: Assumptions Used in Analysis

- Steady-state flow conditions
- Ideal gas behavior
- Negligible heat transfer in pipeline
- Fully developed flow
- Uniform pipe roughness
- No leakage losses
- Constant compressor efficiency